

Module 1: Introduction

1.1 Definition & Concepts

1. Define the terms: Hazard, Vulnerability, Risk **(Frequency: 2, Marks: 5)**
2. Define the terms: Mitigation, Triage **(Frequency: 2, Marks: 5)**
3. Explain Risk and Vulnerability. **(Marks: 5)**
4. Define and differentiate between Risk and Vulnerability in the context of disaster management. **(Marks: 5)**
5. Define the terms disaster, hazard, vulnerability and risk. **(Marks: 5)**
6. Define vulnerability and its significance in disaster management. **(Marks: 4)**

1.2 Effects & Significance

7. Discuss the Direct and indirect effects of disasters. **(Marks: 5)**
8. Analyze the enduring impacts of disasters on affected communities, infrastructure, and socio-economic systems. **(Marks: 5)**
9. Justify the significance of studying Disaster Management, highlighting its role in enhancing resilience, minimizing risks, and promoting sustainable development. **(Marks: 5)**
10. Evaluate the impact of disasters on infrastructure and their hindrance to developmental objects. **(Marks: 6)**
11. Comment on Human repercussions of natural calamities. **(Marks: 5)**

1.3 Global Warming & Climate Change

12. What is Climate Change? What are the effects of Global Warming? **(Frequency: 2, Marks: 5)**
13. Discuss Climate change, focusing on its underlying causes, ecological repercussions, and implications for disaster risk management. **(Marks: 5)**
14. Explain how global warming affects the water cycle and its various components. **(Marks: 5)**
15. Explain global warming and climate change. **(Marks: 6)**

****1.4 Disaster Scenario****

16. What is Disaster Scenario of India? ****(Frequency: 2, Marks: 5)****

17. Provide an overview of various types of Manmade disasters, elucidating their causes, characteristics, and impacts. ****(Marks: 5)****

18. Discuss various types of natural disasters in India and highlight their impacts on life. ****(Marks: 6)****

**Module 2: Natural and Manmade Disasters**

****2.1 Natural Disasters****

19. What are different types of flood? Enlist structural mitigation measures for flood. ****(Frequency: 2, Marks: 10)****

20. Explain the types of landslide and factors affecting them. Give a case study for the same. ****(Marks: 10)****

21. What are droughts? State the mitigation measures taken in drought prone areas. ****(Marks: 10)****

22. Classify different types of droughts and outline structural mitigation measures suitable for drought-prone regions. ****(Marks: 10)****

23. Write a note on Volcanic eruptions. ****(Marks: 8)****

24. Identify and discuss the various hazards which are associated with volcanic eruptions. ****(Marks: 5)****

25. Urban flooding: Implications for major cities. ****(Marks: 5)****

26. Explain disaster management for floods in urban areas. ****(Marks: 5)****

27. Is rapid depletion of ground water a type of disaster? To which category does this belongs? What are the reasons for this problem? ****(Marks: 5)****

****2.2 Manmade Disasters****

28. Explain types of Manmade disasters. ****(Marks: 5)****

29. What are Man-made hazards? Also discuss possibilities of chemical spills.
(Marks: 5)
30. Explain disasters related to industries. Discuss the socioeconomic impact of industrial disasters with a case study. **(Marks: 10)**
31. Discuss the various types of technological disasters and highlight the specific efforts to mitigate such disasters in India. **(Marks: 8)**
32. Comment on radiation hazards. Also discuss possibilities of chemical spills in Mumbai. **(Marks: 6)**
33. Enumerate preparedness measures for minimizing chemical disaster losses.
(Marks: 5)
34. Delve into the paradox of industrialization: economic progress versus the threat of industrial mishaps. **(Marks: 8)**
35. Discuss the role of growing population, industrialization, urbanization, and changing lifestyles in contributing to environmental degradation. What measures can be taken to mitigate their impact? **(Marks: 10)**
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Module 3: Disaster Management, Policy and Administration

3.1 Concepts & Framework

36. Explain PDCA cycle of disaster management. **(Marks: 5)**
37. Examine the multifaceted components of Disaster Management, including preparedness, response, recovery, and mitigation strategies. **(Marks: 10)**
38. Discuss the framework for disaster management in India. **(Marks: 8)**
39. Evolving approaches to disaster management in India. **(Marks: 5)**
40. Write a short note on DM Act 2005. **(Marks: 10)**
41. Discuss in brief the Disaster Management Act 2005. **(Marks: 5)**
42. Write short notes on: National Disaster Management Policy **(Marks: 5)**

3.2 Policy, Command & Coordination

43. Describe the Search and Rescue (SAR) procedure in disaster response, supported by a case study illustrating its implementation and effectiveness. **(Marks: 10)**

44. Explain Triage. Analyze the importance of initial response in disaster management. **(Marks: 8)**

45. Write short notes on: Triage **(Frequency: 2, Marks: 5)**

Module 4: Institutional Framework in India

4.1 Agencies & Responsibilities

46. What are different government agencies responsible for various types of disasters? **(Marks: 5)**

47. Explain roles and responsibilities of NDMA in detail. **(Marks: 5)**

48. Elaborate the guidelines laid down by NDMA for disaster management in India. **(Frequency: 2, Marks: 10)**

49. Examine NDMA's directives for disaster preparedness in India. **(Marks: 6)**

50. What are the scope and responsibilities of NIDM and NDMA? **(Marks: 10)**

51. Explore the scope and responsibilities of the NIDM in disaster preparedness, training, research, and policy formulation. **(Marks: 10)**

52. Outline the functions and duties of NIDM. **(Marks: 6)**

53. Write short notes on: Functions of NIDM **(Frequency: 2, Marks: 5)**

54. Discuss the roles and functions of various government agencies in Disaster Management in India, emphasizing their coordination and collaboration efforts. **(Marks: 10)**

4.2 Technology in DM

55. Discuss the role of GIS and Remote Sensing in disaster management. **(Frequency: 3, Marks: 10)**

56. Explain role of various softwares in disaster management. **(Marks: 10)**

57. Appraise the role of Geo-informatics in disaster management. **(Marks: 10)**

58. Investigate the applications of GIS in disaster management, highlighting its role in spatial analysis, risk assessment, and decision-making. **(Marks: 10)**
59. Appraise the role of GIS and GPS in disaster management. **(Marks: 6)**
60. Explore the applications of GPS and GIS in drought mitigation strategies. **(Marks: 7)**
61. Leveraging technology for disaster preparedness. **(Marks: 5)**
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Module 5: Financing Relief Measures

5.1 Fundraising & NGOs

62. What is role of NGOs in disaster management? Enlist major NGOs working on disaster management. **(Frequency: 3, Marks: 10)**
63. Assess the contributions of various NGOs in Disaster Management, outlining their roles in emergency response, community engagement, and capacity-building initiatives. Additionally, list major NGOs operating globally in this field. **(Marks: 10)**
64. Describe in brief about the role of NGO's in disaster management. **(Marks: 5)**
65. Examine the collaborative efforts of government bodies and NGOs in disaster management. **(Marks: 6)**
66. Explain various means of raising finance for mitigating and managing disasters. **(Frequency: 2, Marks: 10)**
67. Evaluate various fundraising mechanisms for disaster management, considering their effectiveness, sustainability, and ethical implications. **(Marks: 10)**
68. What are the various methods of financial management during a disaster? Explain the government's fund-raising system for disaster response and recovery. **(Marks: 10)**
69. Propose strategies for funding disaster relief efforts and discuss legal considerations. **(Marks: 5)**

5.2 International Aid

70. Investigate the contributions of international aid agencies during extreme crises. **(Marks: 7)**

71. Explain the role of various international agencies for Disaster Management.

(Marks: 6)

72. Global aid organizations in crisis response. **(Marks: 5)**

Module 6: Preventive and Mitigation Measures

6.1 Structural Mitigation

73. Explain Bio shield and Sea wall in detail with schematic diagram. **(Marks: 10)**

74. Define Bioshield and explore its utility in disaster mitigation, particularly in addressing specific hazards such as floods, landslides, and coastal erosion. **(Marks: 5)**

75. Write short notes on: Sea walls, embankments and bio shields **(Marks: 5)**

76. Explain with examples structural and non-structural mitigation measures. **(Marks: 6)**

77. Differentiate between structural and non-structural measures of flood mitigation and discuss the importance of forecasting, warning and monitoring system in India. **(Marks: 8)**

78. Define liquefaction and debate the merits of retrofitting versus relocation in landslide-zone areas. **(Marks: 6)**

79. Explain in detail the design concepts involved in as well as the construction materials used for the safe construction of facilities in case of earthquakes and cyclones. Also discuss the fire resistant facilities that need to be essentially provided in a building/industry. **(Marks: 8)**

6.2 Non-Structural Mitigation

80. What is Community Base Disaster Management (CBDM)? Discuss how it is useful in Indian scenario. **(Marks: 10)**

81. Define Capacity Development and illustrate its application through a case study of Community-Based Disaster Management (CBDM), emphasizing community empowerment, resilience-building, and sustainable development. **(Marks: 10)**

82. Explain the concept of Capacity building along with suitable example. **(Marks: 5)**

83. Write short notes on: Capacity building **(Frequency: 2, Marks: 5)**
84. Write short notes on: Community based disaster preparedness **(Frequency: 2, Marks: 5)**
85. Explain Community based disaster preparedness. **(Marks: 5)**
86. Highlight the protective role of natural ecosystems against disasters, with examples. **(Marks: 6)**
87. What are different mitigation measures for Tsunami and Cyclones? **(Marks: 10)**
88. Explain in detail, vulnerability, with reference to floods and cyclones. List down the preparatory measures for minimizing vulnerabilities related to Tsunami. **(Marks: 6)**
89. Define early warning systems and illustrate their benefits through recent instances of flood, cyclone, and tsunami alerts. **(Marks: 8)**

****6.3 Do's and Don'ts****

90. What are Do's and don'ts in Earthquake, Tsunami and Cyclone? **(Marks: 10)**
91. Enumerate essential Do's and Don'ts for disaster preparedness and response, providing practical guidelines for individuals and communities to mitigate risks and ensure safety. **(Marks: 10)**
92. What are the dos and don'ts in the event of a Covid like pandemic? **(Marks: 6)**
93. Describe non-structural measures for pandemic containment. **(Marks: 6)**
94. Guidelines for earthquake safety. **(Marks: 5)**